

Lecture 17 - Economics and Business Administration

Activity-Based Costing (ABC) Systems

Agenda

- Traditional Systems and Activity-Based Costing (ABC)
- How ABC Systems Work
- Objectives of ABC Systems
- Steps for Implementing ABC
- Other Uses of ABC Systems
- Limitations of ABC

Indirect Cost Allocation Strategies in CMS

Traditional CMS

- Traditional CMS define the methods for determining the full manufacturing cost of a product.
- They therefore focus primarily on production costs, treating non-production costs using simplified methodologies.
- In very simple businesses (few cost objects) and when the majority of costs are direct materials (DM) and direct labour (DL), manufacturing overhead is combined into a single cost pool and allocated using a single allocation base (or cost driver).
- As complexity increases, indirect costs are split into pools defined by process stages (operating departments) and allocated to cost objects using specific drivers.

Activity-Based Costing (ABC)

- ABC Systems measure **all costs relevant to management** and extend cost allocation **beyond production**, including design, marketing, order processing costs, etc.
- In **highly complex situations** (many cost objects) and when **indirect costs represent a high proportion of total costs**, they are first allocated to **individual activities** within a department, function, plant, etc., and **then allocated to cost objects**.

Product Cost and Other Cost Objects

- The study of CMS has taken the unit of product as the cost object.
- Activity-Based Costing (ABC) was developed for two purposes:
  - **To improve product cost measurement;**
  - To improve cost allocation (especially for NON-production costs) to cost objects other than the unit of product.

Why does the controller **need to allocate NON-production costs** and assign costs to objects other than the product?

Suppose the controller wants to know the profitability of a customer. The customer buys many different products and consumes overhead activities (e.g. *accounting staff who must issue, record and collect invoices*, or *the credit recovery unit*).

Example

- Customer Mario Rossi, in the period under review, generated revenues of €10,000,
- purchasing:
  - 100 units of product A — Full Manufacturing Cost (FMC) = €20
  - 50 units of product B — FMC = €60
  - 80 units of product C — FMC = €40

Knowing the unit product cost, the company can easily determine the Cost of Goods Sold:

$$100 \times €20 + 50 \times €60 + 80 \times €40 = €8,200$$

Revenues	**10.000**
Cost of goods sold	**8.200**
Gross Margin	**1.800**

If the company did not consider the **burden imposed by the customer on NON-production costs**, it would not have a reliable measure of **profitability**.

Types of Allocation

The controller must therefore address cost allocation in two distinct but equally fundamental areas:

Allocation of indirect costs to the single product

For this purpose the controller considers **only production costs**, with the objective of determining the unit **FMC**, which in turn into the Cost of Goods Sold

Allocation of indirect costs to other cost objects

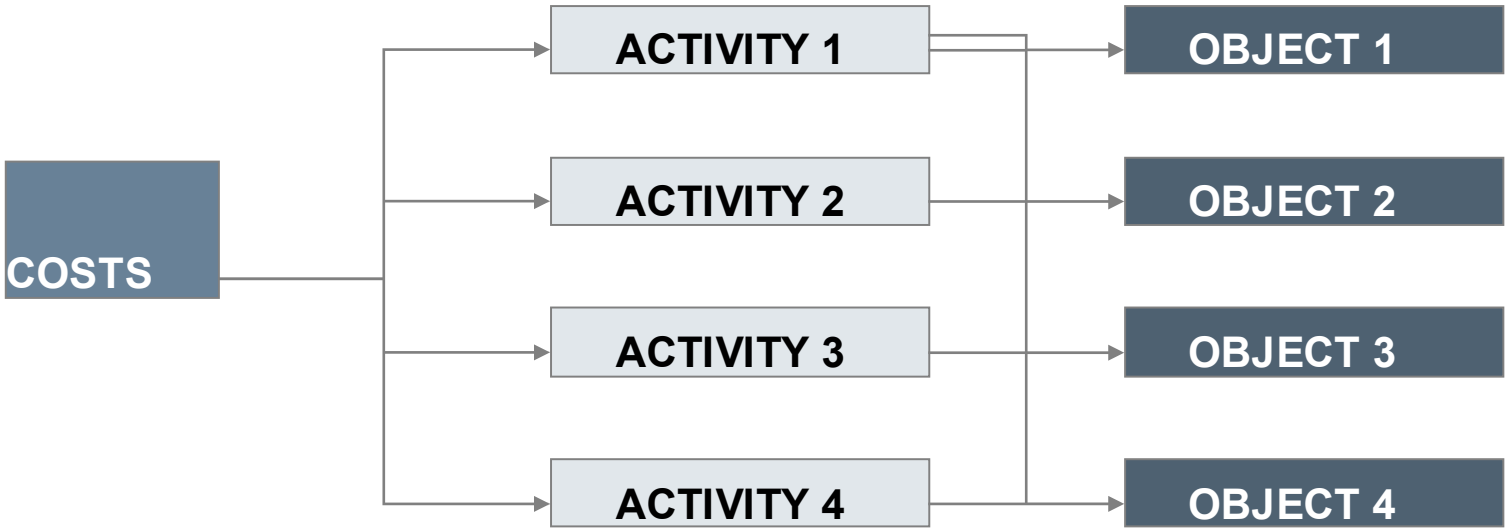
When the controller wants to build income statements (by customer, geographic area, business manager, etc.).
Such income statements often follow responsibility accounting logic and require cost allocations that mainly involve NON-production costs

The Activity-Based Costing (ABC) methodology can support both allocation processes …
… but the different purposes must be kept distinct

How ABC Systems Work

The application of ABC Systems typically involves two steps:

1. **Resource costs** are allocated to **activities** through **resource cost drivers** (usually the % of staff time dedicated to each activity, but also other cost drivers);
2. **Activity costs** are then allocated to **cost objects** (products/services or other objects) through **activity cost drivers** that express the consumption of each activity by those objects (specific drivers for each activity).



Step 1
(Resource cost drivers)

Step 2
(Activity cost drivers)

Exercise: ABC

- ABC S.p.A. uses a traditional single-base cost system. The cost driver for manufacturing overhead allocation is **DL cost.**
- The following information is provided by the company's CMS for component AK25, of which the company produces 10,000 units:

	Total company values	Values for 10,000 units of component AK25	Unit values for component AK25
Direct materials	250,000	25,000	2.5
Direct labour	800,000	60,000	6.0
Manufacturing overhead (150% of DL cost)	1,200,000	90,000	9.0
Total cost	***2,250,000***	***175,000***	***17.5***
Selling price			20.0
Gross operating margin			**2.5**
Margin % on price			**12.5%**

ABC Exercise

- The controller intends to implement a two-step ABC system for more accurate allocation of manufacturing overhead.
- The following activities are identified and the following information is collected:

Manufacturing overhead activities	% time dedicated to each activity	Activity cost driver	Total activity driver value	Total activity driver value used for the production of 10.000 units of the component AK25
Quality control	16%	no. of inspections	2.000	200
Production scheduling	8%	No. of set-ups	1.000	120
Machine set-up	11%	No. of set-ups	1.000	120
Shipping administration	15%	No. of shipments	3.000	300
Production	50%	Machine hours	20.000	2.000
	100%			

ABC Exercise: Solution

Step 1

Allocation of costs to activities (resource driver = % of time dedicated)

Manufacturing overhead activities	% time dedicated to each activity	Indirect costs allocated to activities
Quality control	16%	192,000
Production scheduling	8%	96,000
Machine set-up	11%	132,000
Shipping administration	15%	180,000
Production	50%	600,000
	100%	**1,200,000**

ABC Exercise: Solution

Step 2

Allocation of costs to Component AK25 (activity-specific cost drivers)

Manufacturing overhead activities	Indirect costs allocated to activities	Activity cost driver	Total activity driver value	Cost per unit of driver
Quality control	192,000	no. of inspections	2,000	96.00
Production scheduling	96,000	No. of set-ups	1,000	96.00
Machine set-up	132,000	No. of set-ups	1,000	132.00
Shipping administration	180,000	No. of shipments	3,000	60.00
Production	600,000	Machine hours	20,000	30.00
	1,200,000			

Manufacturing overhead activities	Cost per unit of driver	Activity driver value used for 10,000 units of component AK25	Costs allocated to 10,000 components
Quality control	96.00	200	19,200.0
Production scheduling	96.00	120	11,520.0
Machine set-up	132.00	120	15,840.0
Shipping administration	60.00	300	18,000.0
Production	30.00	2,000	60,000.0
			124,560

ABC Exercise: Income Statement

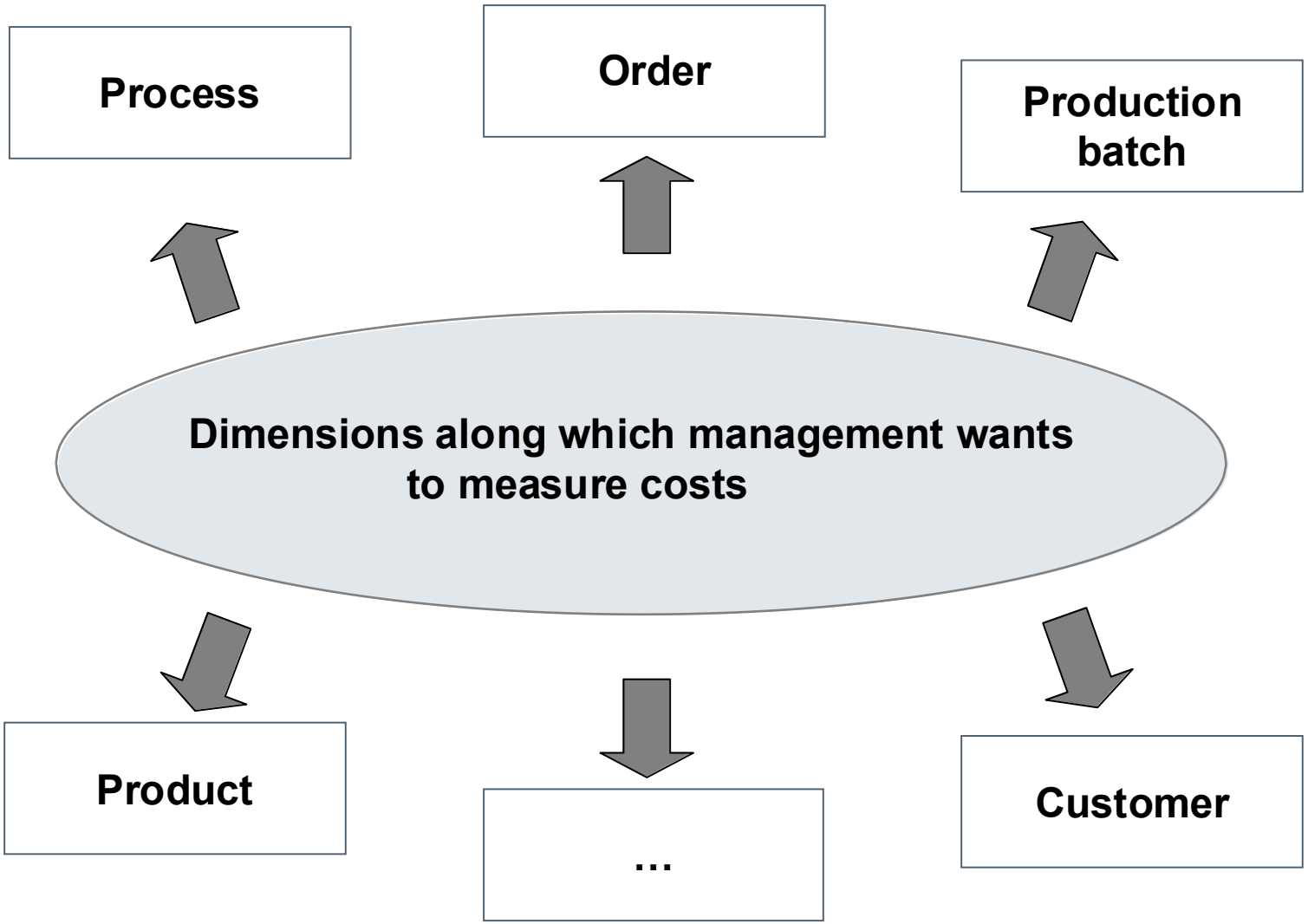
Income Statement using the ABC methodology

	Values for 10,000 units of component AK25	Unit values for component AK25
Direct materials	25,000	2.5
Direct labour	60,000	6.0
Manufacturing overhead (150% of DL cost)	124,560	12.5
Total cost	***209,560***	***21.0***
Selling price		20.0
Gross operating margin		**- 1.0**
Margin % on price		**-5%**

Objectives of ABC Systems

- **Product and service cost measurement**
  - ABC systems provide **more accurate cost information**; by using **multiple cost drivers for different activities**, they allocate more costs to the products that consume more of those activities.
- **Process and activity analysis**
  - ABC **systems allow management to understand the cost of activities performed by different business functions**.
  - ABC systems **make it possible to assess the magnitude of costs relative to the value generated by activities** for the company and its customers.
- **Measurement of customer profitability and other cost objects**
  - ABC systems facilitate the preparation of income statements not only by product but also by **customer**. The unit cost of the activity driver can more readily be applied to assign costs across multiple cost objects (customers, geographic areas, market segments, etc.).

Possible Cost Objects in the Allocation Process



Steps for Implementing an ABC System

1. Identify the **key components of the system** (cost objects, activities, cost drivers);

2. Determine the **relationships between objects**, **activities** and **resources** (interviews with management; ABC requires a deep understanding of business processes);

3. Collect **relevant data on costs and cost drivers**;

4. Calculate and interpret the new cost information.

Other Uses of ABC

Activity-Based Costing (ABC) can be used to identify areas that would benefit from improvements in operational processes.

Information generated by ABC systems can support operational and strategic decision-making

- **ABM (Activity-Based Management) → cost management based on activities.**
- **Costs and decisions → use of cost information in decision-making processes**

Limitations of ABC

ABC systems

are "**operational mechanisms**" that influence behaviour and require incremental resources as well as **significant expertise**.

The **benefit of greater informational relevance must outweigh these additional costs**.

ABC systems

produce "informational" results that differ significantly from those produced by traditional costing systems.

Some managers find it difficult to adapt to this change.

Data derived from the application of the ABC methodology must be used carefully in the decision-making process.

Exercise: Pinco

Pinco S.p.A. produces two products (A and B) in two production departments (D1 and D2). The following information is available for the two products:

	PDTA	PDTB	TOT
Unit direct materials	€ 30,00	€ 10,00	
DL hrs – D1	4	1	
DL hrs – D2	1	4	
Hourly DL rate			€ 20,00
Production volumes	2.000	5.000	

The company records the following manufacturing overhead:

Manufacturing overhead	
Depreciation	500.000
Indirect departmental staff	800.000
Maintenance costs	200.000
Department supervisors	620.000
Indirect materials costs	80.000
	2.200.000

Exercise: Pinco

The controller wants to apply the ABC methodology to allocate manufacturing overhead to the two products. To do so, he identifies the following activities and the following % of staff time dedicated, which he uses as the resource driver:

Activity	% time dedicated
Direct production	50%
Machine set-up	20%
Quality control	15%
Component handling	15%
	100%

He believes that direct production activities can be correctly allocated to products based on DL hours, while for the other activities he collects the following information (total values recorded in the period):

	PDTA	PDTB	TOT
no. of components handled	40.000	60.000	100.000
no. of set-ups	100	100	200
no. of quality inspections	150	50	200

Pinco Exercise: Questions

1. Determine the **unit full manufacturing cost** of both products using **the Activity-Based Costing methodology**
2. Compare the costs allocated under the ABC methodology with those from the other methods and explain the differences

Pinco Exercise: Solutions

1. Pdt A = € 586,89; Pdt B = € 367,24
2. Comparison of Indirect Cost Allocated

	Pdt A	Pdt B
Summary		
Single-base method	314,29	314,29
Process-stage method	296,68	321,33
ABC method	456,89	257,24